EXP.NO-9	Develop an Android application [Calculator]

AIM:

To develop an Android application using controls like Button, TextView, and EditText for designing a calculator that performs basic operations such as Addition, Subtraction, Multiplication, and Division.

ALGORITHM:

1. Open Android Studio and create a new project using Empty Views Activity or Activity.
2. Design the user interface using:
   a. EditText for input numbers
   b. Button for operations (+, −, ×, ÷)
   c. TextView to display the result
3. Assign unique IDs to all UI components in the XML layout file.
4. Open the main activity file (Java/Kotlin).
5. Link UI components using findViewById() (XML) or state variables (Compose).
6. Write logic for:
   a. Addition
   b. Subtraction
   c. Multiplication
   d. Division
7. Handle special cases like:
   a. Empty input
   b. Division by zero
8. Display the result in the TextView.
9. Run the application using an emulator or real device. 10. Test all operations and verify outputs.

CODE:

XML Layout (activity_main.xml)

XML

```
package com.example.calculator

import android.os.Bundle

import android.widget.Button

import android.widget.EditText

import android.widget.TextView
```

```kotlin
import androidx.appcompat.app.AppCompatActivity


class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main) // Make sure your XML file is named activity_main.xml


        val num1 = findViewById<EditText>(R.id.num1)
        val num2 = findViewById<EditText>(R.id.num2)
        val result = findViewById<TextView>(R.id.result)


        val addBtn = findViewById<Button>(R.id.add)
        val subBtn = findViewById<Button>(R.id.sub)
        val mulBtn = findViewById<Button>(R.id.mul)
        val divBtn = findViewById<Button>(R.id.div)


        addBtn.setOnClickListener {
            val n1 = num1.text.toString().toDoubleOrNull() ?: 0.0
            val n2 = num2.text.toString().toDoubleOrNull() ?: 0.0
            result.text = "Result: ${n1 + n2}"
        }


        subBtn.setOnClickListener {
            val n1 = num1.text.toString().toDoubleOrNull() ?: 0.0
            val n2 = num2.text.toString().toDoubleOrNull() ?: 0.0
            result.text = "Result: ${n1 - n2}"
        }


        mulBtn.setOnClickListener {
            val n1 = num1.text.toString().toDoubleOrNull() ?: 0.0
```

```
            val n2 = num2.text.toString().toDoubleOrNull() ?: 0.0
            result.text = "Result: ${n1 * n2}"
        }

        divBtn.setOnClickListener {
            val n1 = num1.text.toString().toDoubleOrNull() ?: 0.0
            val n2 = num2.text.toString().toDoubleOrNull() ?: 0.0
            if (n2 != 0.0) {
                result.text = "Result: ${n1 / n2}"
            } else {
                result.text = "Error: Division by zero"
            }
        }
    }
}
```

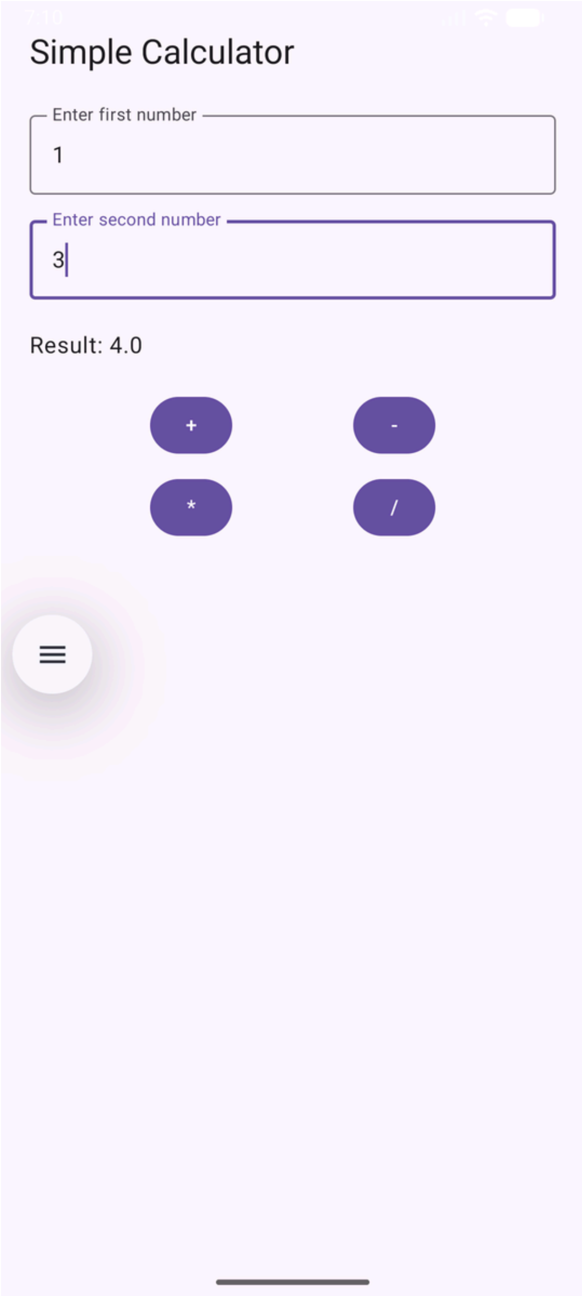
Java Code (MainActivity.java)

```
package com.example.calculator;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText num1, num2;
    TextView result;
    Button add, sub, mul, div;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main); // Ensure your XML file is named activity_main.xml

    num1 = findViewById(R.id.num1);
    num2 = findViewById(R.id.num2);
    result = findViewById(R.id.result);

    add = findViewById(R.id.add);
    sub = findViewById(R.id.sub);
    mul = findViewById(R.id.mul);
    div = findViewById(R.id.div);

    add.setOnClickListener(v -> calculate("+"));
    sub.setOnClickListener(v -> calculate("-"));
    mul.setOnClickListener(v -> calculate("*"));
    div.setOnClickListener(v -> calculate("/"));
}

private void calculate(String op) {
    String val1 = num1.getText().toString();
    String val2 = num2.getText().toString();

    if (val1.isEmpty() || val2.isEmpty()) {
        result.setText("Enter values");
        return;
    }
```

```
        double n1 = Double.parseDouble(val1);
        double n2 = Double.parseDouble(val2);
        double res;

        switch (op) {
            case "+":
                res = n1 + n2;
                break;
            case "-":
                res = n1 - n2;
                break;
            case "*":
                res = n1 * n2;
                break;
            case "/":
                if (n2 == 0) {
                    result.setText("Cannot divide by zero");
                    return;
                }
                res = n1 / n2;
                break;
            default:
                res = 0;
        }

        result.setText("Result: " + res);
    }
}
```

OUTPUT:



RESULT:

Thus, an Android application using Button, TextView, and EditText was successfully developed